

About the Collatz conjecture

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Abstract. This paper refers to the Collatz conjecture. The origin and the formalization of the Collatz problem are presented in the first section, named “Introduction”.

In the second section, entitled “Properties of the Collatz function”, we treat mainly the bijectivity of the Collatz function. Using the obtained results, we construct a (set of) binary tree(s) which “simulate(s)” - in a way that will be specified - the computations of the values of the Collatz function.

In the third section, we give an “efficient” algorithm for computing the number of iterations (recursive calls) of the Collatz function. A comparison between our algorithm and the standard one is also presented, the first being at least 2.25 “faster” (3.00 in medium).

Finally, we describe a class of natural numbers for which the conjecture is true.

1 Introduction

The exact origin of the Collatz conjecture, also called “Syracuse conjecture” or “ $3x + 1$ problem”, is not clearly known. It had orally circulated among the mathematical community for many years. This problem is credited to Lothar Collatz (University of Hamburg). J.H. Conway [1] and R.K. Guy [3] are other important names who studied the problem.

This conjecture asserts that the following program, terminates for any given integer m :

```
n := m;  
while n > 1 do  
  if (n is even) then  
    n := n/2
```

```

else
  n := 3n + 1
endif
endwhile;

```

□

Remark 1.1

1. This represents a useful example when we speak about terminating / nonterminating algorithms or about non-total recursive functions.
2. The above algorithm obviously terminates for $m \leq 1$ (the body of the “while” loop is never executed) and presents no scientific interest. Much more, if $m > 1$, the only way we can have a finite execution is to “reach” the value 1. □

Nabuo Yoneda (University of Tokio) has checked it on a computer for all $n \leq 2^{40}$ and has obtained an affirmative answer. The behaviour of the Collatz “series” seems to be structureless and “random”. For instance, for $n = 2^{500} - 1$, integers greater than $2^{500} \times 10^{88}$ are reached during the computation.

The conjecture may be rephrased as follows:

Conjecture 1 *Let $f : \mathbb{N}^* \rightarrow \mathbb{N}^*$ be the (“Collatz”) function*

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \equiv 0 \pmod{2} \\ 3n + 1 & \text{otherwise} \end{cases}$$

Show that, for any $n \in \mathbb{N}^$, there exists $k \in \mathbb{N}^*$ such that $f^{(k)}(n) = 1$, ($f^{(k)}$ denotes $\underbrace{f \circ \dots \circ f}_{k \text{ times}}$).* □

From now on, we shall denote the Collatz function by **Cf**.

2 Properties of Cf

Theorem 2.1 *Cf is surjective, but not injective.*

Proof. We will show that the equation:

$$(1) \quad \text{Cf}(x) = y$$

has a solution for each y ; moreover, this is not always unique.

Let y be an arbitrary positive integer.

- if x is even, then we have $\text{Cf}(x) = \frac{x}{2}$, that is we get the solution $x = 2y \in \mathbb{N}^*$;
- if x is odd, then there exists $k \in \mathbb{N}$ such that $x = 2k + 1$; (1) becomes $y = \text{Cf}(x) = 3x + 1 = 6k + 4$. Thus, we get $k = \frac{y-4}{6}$ and $x = \frac{y-1}{3}$, from which we immediately deduce that $x \in \mathbb{N}^*$ (because $y = 6k + 4$).

So, it follows that the function Cf is surjective. Moreover, if $y = 6k + 4$ then (1) has exactly two solutions, $x_1 = 2y$ and $x_2 = \frac{y-1}{3}$. Thus Cf is not injective. $\square$

Example 2.1 It is obvious that $Cf(20) = Cf(3) = 10$. $\square$

According to the proof of Theorem 2.1, we can associate for any $y \in \mathbf{N}^*$, $k \in \mathbf{N}^*$ a finite binary tree, with root y having k levels, denoted by $A_k(y) = (V, E)$, $V \subseteq \mathbf{N}^*$. If the tree is not finite, we denoted it by $A(y) = (V, E)$, $V \subseteq \mathbf{N}^*$. For $A(1)$ the set of nodes $(V(1))$ will coincide with the set of **all the natural numbers z for which there exists an $i \in \mathbf{N}^*$ such that $f^{(i)}(z) = 1$** . The Collatz conjecture may now be rephrased as:

$$\mathbf{N}^* = V(1).$$

In fact, any $A(y)$ is a subtree in $A(1)$ and the trees $A_k(y)$ and $A(y)$ “simulate” the computation of the values of Cf .

Example 2.2

1. Following Example 2.1, the corresponding tree for $\langle 10, 1 \rangle$ has two levels (see Fig. 1a).
2. The corresponding tree associated to $\langle 9, 2 \rangle$ is presented in Fig. 1b.

Example 2.3 The first 8 levels for $A(1)$ are presented in Fig. 1c.

The dotted lines represent a circuit that has been ignored. $\square$

In the following, we present an alternative method for generating all the trees $A_k(1)$.

Algorithm 1

Input: Cf and $k \in \mathbf{N}^*$.

Output: $A_k(1)$.

Method:

```

begin
   $Level_0 := \{1\}; V := \{1\}; E := \emptyset;$ 
  for  $i := 1$  to  $k$  do begin
     $Level_i := \emptyset;$ 
    for each  $y \in Level_{i-1}$  do begin
       $x := 2 * y;$ 
      push_tree( $x$ );
      if  $y \equiv 4 \pmod{6}$  then begin
         $x \equiv \frac{y-1}{3};$ 
        push_tree( $x$ )
      endif
    endfor
  endfor
end;

```

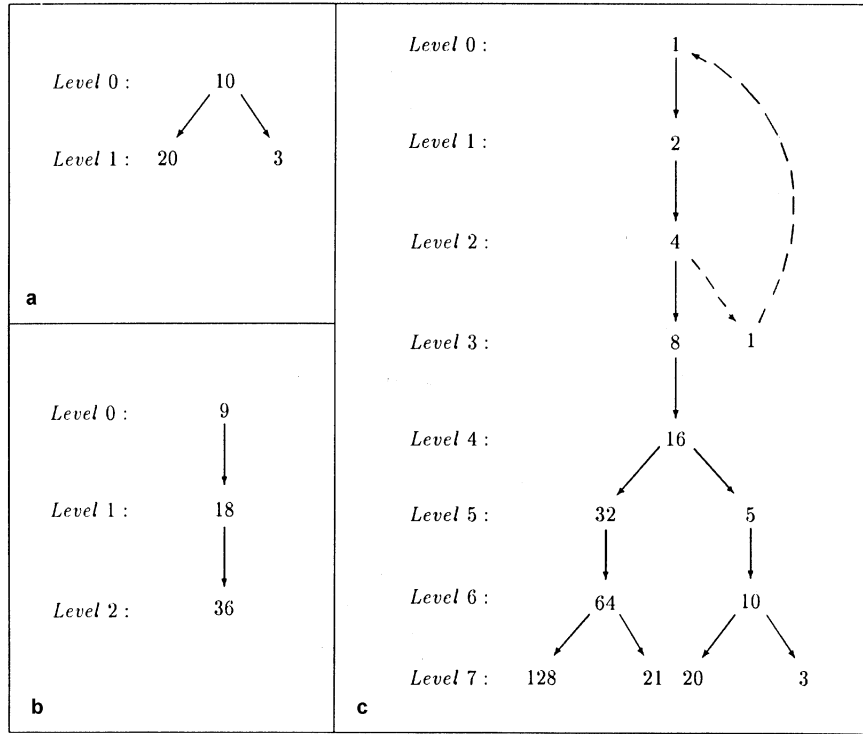


Fig. 1a-c.

where:

```

procedure push_tree(x);
begin
  if  $x \notin V$  then begin
     $V := V \cup \{x\};$ 
     $Level_i := Level_i \cup \{x\};$ 
     $E := E \cup \{(y, x)\};$ 
  endif
end;

```

□

Theorem 2.2 (correctness of the Algorithm 1)

The Algorithm 1 generates the trees $A_k(1)$ for all $k \in \mathbb{N}^*$.

Proof. According to Theorem 2.1 and Remark 2.1, using induction by i , it is obvious that:

$$(\forall i) (z \in Level_i \iff f^{(i)}(z) = 1) \square$$

Example 2.4 It is easily seen (Example 2.3) that $10 \in Level_6$ in any tree having at least 7 levels and that $f^{(6)}(10) = f(f(f(f(f(f(10)))))) = 1$. □

Using the computer we have deduced all the numbers (vertices) of the tree $A(1)$ until the level 31. **The conjecture is still true.**

3 An “efficient” algorithm for computing k

The results presented in this section are the basis for our original conclusions.

Notation 3.1 Following the above algorithms and the definition of $\mathbf{Cf}$, we are sistematicaly using the “transformation rules”:

- (a) $x \rightarrow \frac{x}{2}$;
- (b) $x \rightarrow 3x + 1$. □

Lemma 3.1 For each $p \in \mathbf{N}$ and odd $r \in \mathbf{N}$, we have:

$$f^{(2k)}(2^p \cdot r + 2^p - 1) = 3^k(2^{p-k} \cdot r + 2^{p-k} - 1) + 3^k - 1, \forall k \in \mathbf{N}, 0 \leq k \leq p.$$

Proof. If r is not odd (i.e. $r = 2^q \cdot s$, s odd), we may rewrite $2^p \cdot r$ by $2^{p+q} \cdot s$. We proceed by induction on k .

Basis: $k = 0$. Obvious.

Inductive step: We suppose the affirmation true for k , and we prove it for $k + 1$. Using the inductive hypothesis, we get $f^{(2k+2)}(2^p \cdot r + 2^p - 1) = f^{(2)}(f^{(2k)}(2^p \cdot r + 2^p - 1)) = f^{(2)}(3^k(2^{p-k} \cdot r + 2^{p-k} - 1) + 3^k - 1)$.

Clearly $k < p$, so 2^{p-k} is even. Therefore $2^{p-k} \cdot r + 2^{p-k} - 1$ is odd. Thus $3^k(2^{p-k} \cdot r + 2^{p-k} - 1)$ is odd. But $3^k - 1 \equiv 0 \pmod{2}$, because $3^k - 1 = 2(3^{k-1} + 3^{k-2} + \dots + 1)$ for $k \geq 1$ and $3^k - 1 = 0$ for $k = 0$. Therefore $3^k(2^{p-k} \cdot r + 2^{p-k} - 1) + 3^k - 1$ is odd and the rule (b) must be applied. Hence:

$$\begin{aligned} f^{(2)}(3^k(2^{p-k} \cdot r + 2^{p-k} - 1) + 3^k - 1) &\stackrel{(b)}{=} f(3[3^k(2^{p-k} \cdot r + 2^{p-k} - 1) + 3^k - 1] + 1) = \\ &= f(3^{k+1} \cdot 2^{p-k} \cdot r + 3^{k+1} \cdot 2^{p-k} - 3 + 1) \stackrel{(a)}{=} 3^{k+1} \cdot 2^{p-k-1} \cdot r + 3^{k+1} \cdot 2^{p-k-1} - 1 = \\ &= 3^{k+1}(2^{p-k-1} \cdot r + 2^{p-k-1} - 1) + 3^{k+1} - 1. \end{aligned} \quad \square$$

Theorem 3.1 $f^{(2p)}(2^p \cdot r + 2^p - 1) = 3^p \cdot r + 3^p - 1, \forall p, r \in \mathbf{N}$.

Proof. We take $k = p$ in Lemma 3.1. □

Remark 3.1 If a number of the form $2^p \cdot r + 2^p - 1$ ($p \in \mathbf{N}$, r odd) is a vertex in a tree $A(y)$, then the greater one $3^p \cdot r + 3^p - 1$ is also a node in $A(y)$ (but on a “previous” level). Thus, the number of steps needed to test if a number z is in $A(1)$ can be - sometimes - significantly reduced. □

Corrolary 3.1 Let $p \in \mathbf{N}$ and odd $r \in \mathbf{N}$ be any two numbers. Then:

- a) If there exists $k \in \mathbf{N}$ such that $f^{(k)}(3^p \cdot r + 3^p - 1) = 1$ then $f^{(k+2p)}(2^p \cdot r + 2^p - 1) = 1$.
- b) If there exists $k \in \mathbf{N}, k > 2p$ such that $f^{(k)}(2^p \cdot r + 2^p - 1) = 1$ then $f^{(k-2p)}(3^p \cdot r + 3^p - 1) = 1$.

Proof. a) From Theorem 3.1, we have $f^{(k+2p)}(2^p \cdot r + 2^p - 1) = f^{(k)}(f^{(2p)}(2^p \cdot r + 2^p - 1)) = f^{(k)}(3^p \cdot r + 3^p - 1) = 1$.

b) Similarly as above. □

Remark 3.2 From the Theorems 2.2 and 3.1, we can immediately deduce that in any tree $A_k(y)$ we have ($\forall i, 2 \leq i \leq k$):

1. If the number $2r + 1$ is on the level i , then the number $3r + 2$ is on the level $i - 2$ ($f^{(2)}(2^1 \cdot r + 2 - 1) = 3^1 \cdot r + 3 - 1$).
2. If $2r$ is on the level i , then r is on the level $i - 1$.
3. If $4r + 3$ is on the level i , then $9r + 8$ is on the level $i - 4$.
4. If $4r$ is on the level i , then r is on the level $i - 2$.
5. If $8r + 7$ is on the level i , then $27r + 26$ is on the level $i - 6$.
6. If $8r$ is on the level i , then r is on the level $i - 3$.

Other similar facts can also be emphasized. $\square$

It is obvious that if $n \in \mathbf{N}^*$ is a vertex in $A(1)$ on the level $k(=k(n))$, the Algorithm 1 needs k iterations (executions of the body of the “for” loop and **concerning the number of successive transformations of n , that is recursive calls of Cf**) to “discover” this fact. By using our last results, we can immediately develop a “faster” algorithm (trivially derived from Algorithm 1, called Algorithm 1’ and not explicitly listed here) which “needs” only $k_1 \leq k$ iterations; $k_1(=k_1(n))$ is “almost always” less than k . The following algorithm **directly** computes **both** k and k_1 (also “embodying” the Algorithm 1’) and it eventually stops for any $n \in \mathbf{N}^*$ for which there exists an i such that $f^{(i)}(n) = 1$ (below, $k \equiv niv$, $k_1 \equiv pas$).

Algorithm 2

Input: $n \in \mathbf{N}^*$.

Output: niv (such that $\mathbf{Cf}^{niv}(n) = 1$ following Algorithm 1) and pas (the maximum number of iterations in Algorithm 1’).

Method:

```

begin
   $m := n$ ;  $pas := 0$ ;  $niv := 0$ ;
  while1 ( $m > 1$ ) do begin
     $pas := pas + 1$ ;  $k2 := 1$ ;  $k := 2$ ;  $modul := 1$ ;
    while2 ( $m \equiv modul \pmod{k}$ ) do begin
       $k := 2 * k$ ;
       $niv := niv + 2$ ;
       $modul := k - 1$ ;
       $k2 := k2 * 3$ ;
    endwhile2;
    while3 ( $m \equiv 0 \pmod{k}$ ) do begin
       $k := 2 * k$ ;
       $niv := niv + 1$ ;
       $k2 := k2 * 3$ ;
    endwhile3;
     $k := trunc(k/2)$ ;
     $p := m \div k$ ;
     $p1 := k2 * p + k2 - 1$ ;
    if ( $m \not\equiv 0 \pmod{k}$ ) then  $m := p1$ 
  endwhile1

```

```

    else  $m := p$ ;
  endwhile1;
end

```

□

Before presenting a formal proof of the correctness of the Algorithm 2, an example may be useful.

Example 3.1 Let $n = 23$. We shall compute and compare k_1 and k . For $n = 23$ in Algorithm 2 we start with the following values in the “while1” loop: $k_2 = 1$, $pas = 1$, $k = 2$, $modul = 1$. Let us examine the successive iterations for the “while2” loop:

- $23 \equiv 1 \pmod{2}$? Yes.
 - $k = 4$
 - $niv = 2$
 - $modul = 3$
 - $k_2 = 3$
- $23 \equiv 3 \pmod{4}$? Yes.
 - $k = 8$
 - $niv = 4$
 - $modul = 7$
 - $k_2 = 9$
- $23 \equiv 7 \pmod{8}$? Yes.
 - $k = 16$
 - $niv = 6$
 - $modul = 15$
 - $k_2 = 27$
- $23 \equiv 15 \pmod{16}$? No.

So the **while2** loop ends after 3 steps and we have obtained $k = 8$, $p = 2$, $p_1 = 80$. Therefore (we hope that the chosen notation is self-explanatory):

$$niv(23) = 6 + niv(80)$$

Now the “while1” loop starts with $m = 80$, and so on. We then successively get:

$$niv(80) = 4 + niv(5)$$

$$niv(5) = 2 + niv(8)$$

$$niv(8) = 3 + niv(1)$$

At this step, we have $m = 1$, $pas = 4$, $k_2 = 4$ and the “while1” loop also ends. Hence $niv(1) = 0$ (by convention) and we finally get $niv(23) = 6+4+2+3+0 = 15$. We may say that Algorithm 2 is $\frac{15}{4} = 3.75$ times “faster” than Algorithm 1. In fact, this “fastening” rate is not a real one because the iterations of the Algorithm 2 “hide” other concrete computation steps. □

Definition 3.1 Let $n \in \mathbf{N}^*$, niv and pas be the input and respectively the outputs of the Algorithm 2. Then $R_{acc}(n) = \frac{niv}{pas}$ is called the **acceleration ratio** and $R_{acc} : \mathbf{N}^* \rightarrow \mathbf{R}$ - the **acceleration function**. □

Theorem 3.2 *The following relations hold:*

- (1) $R_{acc}(n) \geq 1, 5, \forall n > 2;$
- (2) $R_{acc}(n) \leq i, \forall n, 2^i \leq n < 2^{i+1}, \forall i \geq 2.$

Proof. (1) The following cases must be considered for the given n :

- (1.1) $n = 4m;$
- (1.2) $n = 4m + 1;$
- (1.3) $n = 4m + 2;$
- (1.4) $n = 4m + 3;$

From Remark 3.2 we correspondingly deduce the following identities for the above mentioned situations:

- (1.1) $R_{acc}(n) = 2;$
- (1.4) $R_{acc}(n) = 4;$

As for the other two cases:

(1.2) $4m + 1 = 2(2m) + 1 \xrightarrow{*} 6m + 2$ (see Notation 3.1) in one “step” for Algorithm 2 (or 1’), and in two steps for Algorithm 1; for $6m + 2$ we obtain $3m + 1$ in one step for both algorithms; so $R_{acc}(4m + 1) = \frac{3}{2} = 1, 5;$

(1.3) $4m + 2 = 2(2m + 1) \xrightarrow{(a)} 2m + 1$, in one step for both algorithms; for $2m + 1$ we get $3m + 2$ in one step for Algorithm 2, and two steps for Algorithm 1; hence $R_{acc}(4m + 1) = \frac{3}{2} = 1, 5.$

Therefore $R_{acc}(n) \geq 1, 5, \forall n \in \mathbf{N}^*.$

(ii) The most “convenient” case ($n = 2^k$) yields $R_{acc}(n) = k$ (k iterations are also enough for numbers $n, 2^k < n < 2^{k+1}$). $\square$

Example 3.2 In the following, we construct the corresponding tree ($A_{15}(1)$, see Fig. 2.) obtained with Algorithm 1 for the number ($n = 23$) considered in Example 3.1. The boldface numbers placed in circles, point out the corresponding (sub)tree generated by Algorithm 2 (or 2’). $\square$

Remark 3.3 If we suppose that $n = 2^k \cdot m, 2^k \cdot m + 1, \dots, 2^k \cdot m + 2^k - 1$ have an equal “occurrence” probability, we may compute the average:

$$P_{med}(n) = \frac{\sum_{i=2^n}^{2^{n+1}-1} R_{acc}(i)}{2^n}$$

For $n = 2$ (Theorem 3.2) we get:

$$P_{med}(2) = \frac{2 + 1, 5 + 1, 5 + 4}{4} = 2, 25.$$

For $n = 3$, we have

$$P_{med}(3) = \frac{3 + 1, (6) + 1, (3) + 2, 5 + 2 + 2 + 2, 5 + 6}{8} = 2, 625$$

and for $n = 4$:

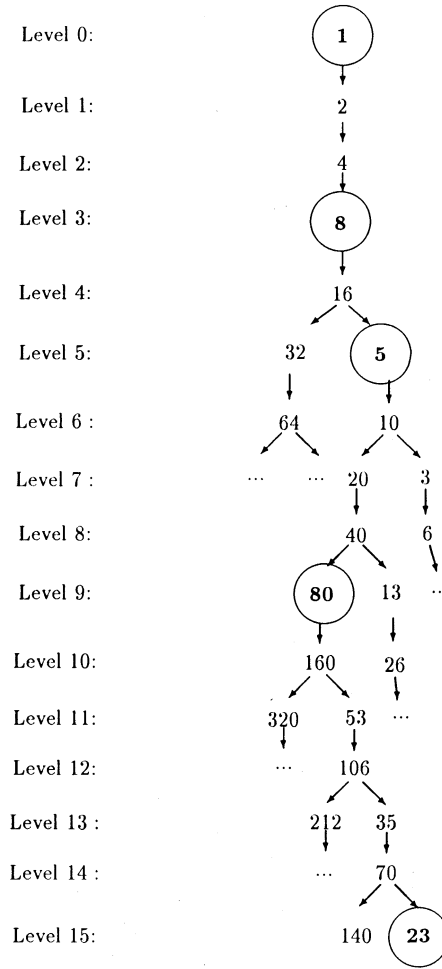


Fig. 2.

$$P_{med}(4) = \frac{4 + 1, 5 + 1, 5 + 3 + 1, (6) + 2, 5 + 2 + 3, 5 + 2, 5 + 2, (3) + 1, (6) + 2, (3) + 3 + 2 + 3, 5 + 8}{16} = 2,8125.$$

Similarly, we deduce:

$$P_{med}(5) = 2.90625, \quad P_{med}(6) = 2.953125 \text{ a.s.o.}$$

If we put $a_n = \sum_{k=2^n}^{2^{n+1}-1} R_{acc}(k)$, we conjecture that the following recurrence relation holds:

$$\begin{cases} a_1 = 3 \\ a_n = 2 \cdot a_{n-1} + 3 \end{cases}$$

Hence we have $a_n = 3 \cdot (2^n - 1)$, and thus

$$\lim_{n \rightarrow \infty} P_{med}(n) = \lim_{n \rightarrow \infty} \frac{a_n}{2^n} = 3 \quad \square$$

Conjecture 2 *The asymptotic limit of the average acceleration ratio equals to 3.*
 $\square$

Theorem 3.3 (correctness of the Algorithm 2) *If $n \in \mathbf{N}^*$ and there exists an $i \in \mathbf{N}$ such that $\mathbf{Cf}^{(i)}(n) = 1$, then Algorithm 2 eventually stops. Moreover, denoting by k the smallest natural number such that $\mathbf{Cf}^{(k)}(n) = 1$ (k equals also the number of iterations - in the known sense - for Algorithm 1), and by k_1 the number of iterations for Algorithm 2 - also in the already fixed sense - we have that $k_1 \leq k$ (the equality holds only for $n = 1, 2$). The operations in “while2” and “while3” loops “correspond” to the computation of the values of the Collatz function.*

Proof.

- The fact that Algorithm 2 eventually stops in the above conditions is immediate: the **while1** loop is never executed for numbers smaller than 1.
- The number of iterations for Algorithm 1 - in constructing $A(1)$ - clearly equals the smallest k such that $f^{(k)}(n) = 1$ (for each $n \in \mathbf{N}^*$ on level $k + 1$ in $A(1)$, supposing that n is not also on a “previous” level).
- For each step (iteration), Algorithm 1 computes - in fact - for the elements on the level i , the elements from the level $i - 1$ (in $A(1)$), applying **only once** the function $\mathbf{Cf}$. In Algorithm 2, we have:
 - if the element $m = 2^p \cdot r$ (p - odd) is on the level i in $A(1)$, then we “**jump**” (using the “inverse” of $\mathbf{Cf}$) directly **to the level** $i - p$, and the current element equals to r ;
 - if the element $m = 2^p \cdot r + 2^p - 1$ (p - odd) is on the level i in $A(1)$, then we “**jump**” **to the level** $i - 2 * p$, and the current element equals to $3^p \cdot r + 3^p - 1$.
- A “CASE statement” in Algorithm 2 may be inserted in a convenient place such as to point out more clearly this affirmation.
- The relation $k_1 \leq k$ directly follows from Theorem 3.2. $\square$

4 The Collatz conjecture - Particular cases

Although we have not proved that the Collatz conjecture holds true for **any** $n \in \mathbf{N}^*$, a class of natural numbers for which the conjecture is true can be derived from the above facts.

The main idea is to start from a “trivial” class of natural numbers for which the conjecture obviously holds. This is $\{n = 2^k \mid k \geq 1\}$.

Lemma 4.1 $f^{(2t+3)}\left(\frac{2^{2t+2}-1}{3}\right) = 1, \forall t \geq 0.$

Proof. Choosing $p = 1$ in Corollary 3.1 we get $f^{(k)}(3r + 2) = f^{(k+2)}(2r + 1)$. Then there exists an r such that $3r + 2 = 2^k$. In fact $r = \frac{2^k - 2}{3} = \frac{2(2^{k-1} - 1)}{3}$. It is clear that we must choose the number k odd ($k = 2t + 1$, $t \in \mathbf{N}$). We obtain $r = \frac{2(2^{2t} - 1)}{3} = 2(2^{2t-2} + 2^{2t-4} + \dots + 1)$. Therefore it follows that $2r + 1 = 2^{2t} + 2^{2t-2} + \dots + 2^2 + 1$; using Theorem 3.1 we obtain:

$$f^{(2t+1)}(2^{2t+1}) = f^{(2t+3)}(2^{2t} + 2^{2t-2} + \dots + 2^2 + 1).$$

But we know that $f^{(2t+1)}(2^{2t+1}) = 1$ and hence

$$f^{(2t+3)}\left(\frac{2^{2t+2} - 1}{3}\right) = 1, \quad \forall t \geq 0$$

□

In the following we try to find *the most general class of natural numbers computed with this method* for which the conjecture is true. Let p be an arbitrary fixed natural number. From now on, we make the following notations:

- we shall denote by $\mathcal{M}_3$ the set $\{3k \mid k \in \mathbf{N}\}$ ($\mathcal{M}_3 - 1 = \{3k + 2 \mid k \in \mathbf{N}\}$ a.s.o.);
- we shall denote by $\perp$ the non-divisibility relation; for example, “3 does not divide n ” will be written as “ $3 \perp n$ ”.

Theorem 4.1 *The following relations hold:*

- a) $3 \perp 2^{2k} + 1, \quad \forall k \in \mathbf{N}$;
- b) $3^{k+1}/2^{3^k} + 1$, but $3^{k+2} \perp 2^{3^k} + 1, \quad \forall k \in \mathbf{N}$;
- c) if p is an odd number which is not divided by 3, then $3^{k+1}/2^{3^k \cdot p} + 1$, but $3^{k+2} \perp 2^{3^k \cdot p} + 1, \quad \forall k \in \mathbf{N}$.

Proof. a) We have $2^{2k} + 1 = 2(2^{2k-1} + 1) - 1 = \mathcal{M}_3 - 1$ and therefore 3 does not divide $2^{2k} + 1$;

b) Let us consider the following predicate:

$$P(k) : \quad 3^{k+1}/2^{3^k} + 1 \text{ and } 3^{k+2} \perp 2^{3^k} + 1, \\ \text{and } 3/2^{3^k \cdot 2} - 2^{3^k} + 1 \text{ and } 9 \perp 2^{3^k \cdot 2} - 2^{3^k} + 1.$$

To prove $P(k)$ - true, we proceed by induction on k .

Basis: For $k = 0$, $k = 1$ - immediate.

Inductive step: Let us suppose that $P(n)$ is true. We must prove that $P(n + 1)$ is true.

$$\text{We have } 2^{3^{n+1}} + 1 = (2^{3^n})^3 + 1 = (2^{3^n} + 1)(2^{3^n \cdot 2} - 2^{3^n} + 1).$$

It remains to show that $3/2^{3^n \cdot 2} - 2^{3^n} + 1$ and $9 \perp 2^{3^n \cdot 2} - 2^{3^n} + 1$. From the inductive hypothesis, we know that there exists an $m \in \mathbf{N}$ such that $2^{3^n \cdot 2} - 2^{3^n} + 1 = 3m$. It results that $2^{3^n \cdot 2} = 2^{3^n} + 3m - 1$. Hence:

$$\begin{aligned} (2^{3^n \cdot 2})^3 &= (2^{3^n} + 3m - 1)^3 \iff \\ \iff (2^{3^{n+1} \cdot 2}) &= 2^{3^{n+1}} + 3(3m - 1) \cdot 2^{3^n \cdot 2} + 3(3m - 1)^2 \cdot 2^{3^n} + (3m - 1)^3 \end{aligned}$$

Therefore $2^{3^{n+1} \cdot 2} - 2^{3^{n+1}} + 1 = 3(3m - 1) \cdot 2^{3^n \cdot 2} + 3(3m - 1)^2 \cdot 2^{3^n} + (3m - 1)^3 + 1$. Thus, we have $3/2^{3^{n+1} \cdot 2} - 2^{3^{n+1}} + 1$, but $9 \nmid 2^{3^{n+1} \cdot 2} - 2^{3^{n+1}} + 1$. Therefore $3^{n+2}/2^{3^{n+1}} + 1$, and $3^{n+3} \nmid 2^{3^{n+1}} + 1$.

c) Let p be an odd number, which is not divided by 3. We have

$$2^{3^k \cdot p} + 1 = (2^{3^k} + 1) \left((2^{3^k})^{p-1} - (2^{3^k})^{p-2} + \dots + 1 \right).$$

Using point b), we know that $3^{k+1}/2^{3^k} + 1$ and $3^{k+2} \nmid 2^{3^k} + 1$. Now we must show that $3 \nmid (2^{3^k})^{p-1} - (2^{3^k})^{p-2} + \dots + 1$.

The affirmation $3 \nmid (2^{3^k})^{p-1} - (2^{3^k})^{p-2} + \dots + 1$ follows much easier since, for any k , $2^{2k} = \mathcal{M}_3 + 1$, $2^{2k+1} = \mathcal{M}_3 - 1$, so $(2^{3^k})^{p-1} - (2^{3^k})^{p-2} + \dots + 1 = \mathcal{M}_3 + p$, which is not divisible by 3.

The point (ii) is proved in an analogous way. $\square$

Theorem 4.2 (some numbers for which the Collatz Conjecture is true)

$$f^{(3^m \cdot n + 2m + 2)} \left(\frac{2^{m+1}(2^{3^m \cdot n} + 1)}{3^{m+1}} - 1 \right) = 1, \quad \forall m \in \mathbf{N}, \quad \forall n \in \{6t + 1, 6t + 5 \mid t \in \mathbf{N}\}.$$

Proof. According to Theorem 3.1 we deduce that

$$(1) \quad f^{(k)}(3^p \cdot r + 3^p - 1) = f^{(k+2p)}(2^p \cdot r + 2^p - 1), \quad \forall p, r \in \mathbf{N}.$$

We may put $3^p \cdot r + 3^p - 1 = 2^k$. Hence $f^{(k)}(3^p \cdot r + 3^p - 1) = f^{(k)}(2^k) = 1$. Now, let us find the numbers p and r .

We have $3^p \cdot (r + 1) = 2^k + 1$; so $r + 1 = \frac{2^k + 1}{3^p}$ and therefore $r = \frac{2^k + 1}{3^p} - 1$. But $r \in \mathbf{N}$, so it results $\frac{2^k + 1}{3^p} \in \mathbf{N}$, that is $3^p \mid 2^k + 1$. We remark that for $p = 1$ and k odd, we get the affirmation of Lemma 4.1.

Now, let p be an arbitrary natural number. According to Theorem 4.1, (point c)), we know that $3^{m+1}/2^{3^m \cdot n} + 1$ and $3^{m+2} \nmid 2^{3^m \cdot n} + 1$, $\forall n \in \{6t + 1, 6t + 5 \mid t \in \mathbf{N}\}$.

Hence, we have $r = \frac{2^{3^m \cdot n} + 1}{3^{m+1}} - 1$. Now, we get $2^p \cdot r + 2^p - 1 = 2^{m+1} \cdot \left(\frac{2^{3^m \cdot n} + 1}{3^{m+1}} - 1 \right) + 2^{m+1} - 1 = \frac{2^{m+1}(2^{3^m \cdot n} + 1)}{3^{m+1}} - 1$, where $n \in \{6t + 1, 6t + 5 \mid t \in \mathbf{N}\}$.

From (1) we have

$$f^{(3^m \cdot n)}(2^{3^m \cdot n}) = f^{(3^m \cdot n + 2m + 2)} \left(\frac{2^{m+1}(2^{3^m \cdot n} + 1)}{3^{m+1}} - 1 \right).$$

But $f^{(3^m \cdot n)}(2^{3^m \cdot n}) = 1$, and therefore $f^{(3^m \cdot n + 2m + 2)} \left(\frac{2^{m+1}(2^{3^m \cdot n} + 1)}{3^{m+1}} - 1 \right) = 1$, $\forall n \in \{6t + 1, 6t + 5 \mid t \in \mathbf{N}\}$.

If we choose $m = 0$, and we consider all the possible cases for which $3/n$, we obtain again the affirmation of Lemma 4.1. $\square$

5 Conclusions

We have deduced that for all the numbers of the form

$$\frac{2^{m+1}(2^{3^m \cdot n} + 1)}{3^{m+1}}, \quad \forall m \in \mathbf{N}, \quad \forall n \in \{6t + 1, 6t + 5 \mid t \in \mathbf{N}\},$$

the Collatz conjecture holds true and Algorithm 1 needs $3^m \cdot n + 2m + 2$ steps to find this.

An “efficient” algorithm for getting this correct answer has been also described. We have presented a comparison between our algorithm and the standard (trivial) one, the first being at least 2.25 “faster” (3.00 in medium). We conjecture that the asymptotic convergence ratio is 3.00. Unfortunately, we do not have the possibility to compare this algorithm with the non-trivial one, but we think that “the novel idea” may be useful.

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